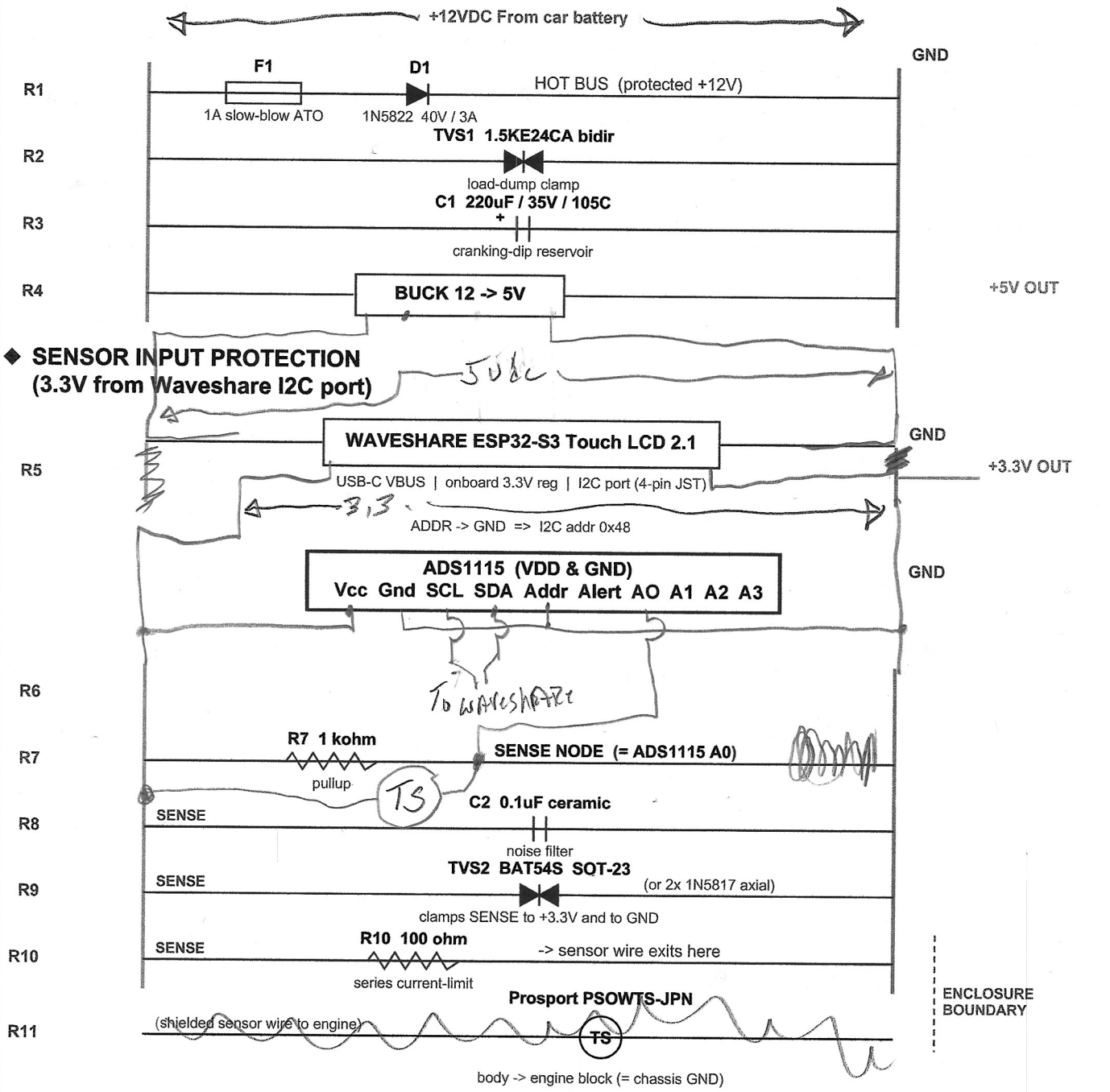
FIREBIRD WATER TEMP GAUGE

Power & Signal Wiring | Ladder Diagram

1967 Pontiac Firebird | ESP32-S3 + ADS1115 + Prosport PSOWTS-JPN | Rev 3 (wiring print)



◆ **+5V -> WAVESHARE ESP32-S3**

+5V

LM2596 / MP1584 1A min

BENCH BUILD R1 jumper D1 until 1N5822 arrives. R2 leave TVS1 pads empty until 1.5KE24CA arrives.

◆ **I2C BUS (4-pin JST on back of Waveshare board)**

R12 SDA ADS1115 SDA

SCL ADS1115 SCL

BILL OF MATERIALS

F1	Fuse 1A slow-blow ATO + inline holder	**D1**	1N5822 Schottky, 40V / 3A axial
TVS1	1.5KE24CA bidirectional (load-dump clamp)	**C1**	220uF 35V 105C electrolytic, radial
TVS2	BAT54S SOT-23 OR 2x 1N5817 axial	**C2**	0.1uF 50V ceramic disc
R7	1 kohm 1% 1/4W metal-film (pullup)	**R10**	100 ohm 1% 1/4W metal-film (limit)
BUCK	12V -> 5V buck module (LM2596 / MP1584)	**TS**	Prosport PSOWTS-JPN coolant temp sender

CHANGELOG

Rev 2 - TVS2 changed from P6KE6.8A to BAT54S / 1N5817. P6KE clamps ~10.5V, too high for ADS1115 (3.6V abs max).
Rev 2.1 - Sender corrected to Prosport PSOWTS-JPN. BENCH notes added for what to do until D1 / TVS1 arrive.
Rev 3 - Ladder redrawn: single rung labels, BENCH notes in callout, changelog broken out of BOM.

STAR-GROUND: all GND connections meet at one point on the protoboard. Keep the long sensor wire shielded; route away from ignition coils.
1N5817 option for R9 (if using 2 axial diodes): D-up anode->SENSE / cathode->+3.3V. D-down anode->GND / cathode->SENSE. Banded end = cathode.